SIMATS ENGINEERING

CO2_ASSESSMENT TOOL3_ Dry Run Evaluation

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

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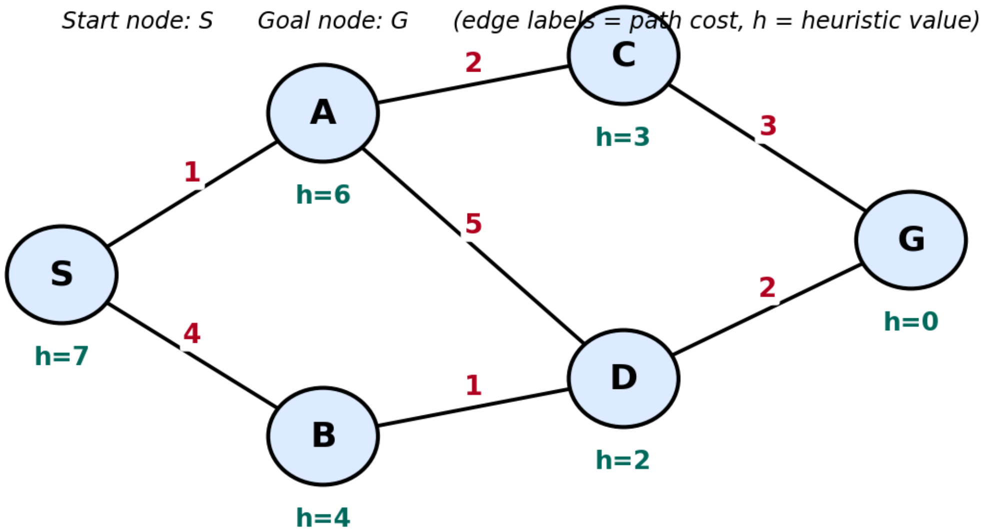
COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

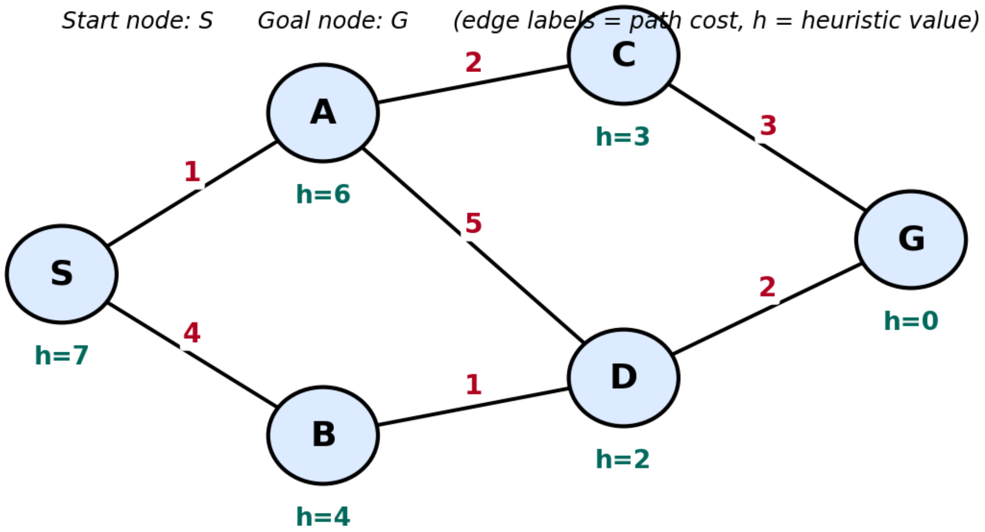
Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:50**

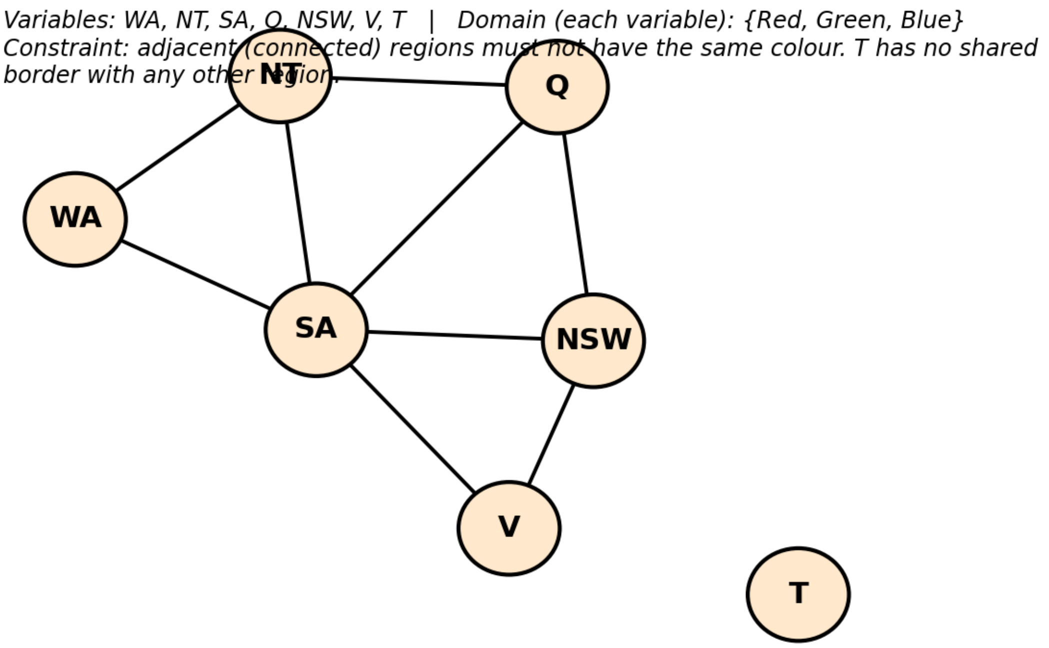
Q1. Consider a suitable state-space graph (with heuristic values h(n) assigned to each node) of your choice for a given problem. Perform a dry run of Greedy Best-First Search from the start node to the goal node. Clearly show the frontier/open list, the node selected for expansion at each step, and the final path obtained.



Q2. For a suitable weighted state-space graph (with edge costs and heuristic values h(n)) of your choice, perform a dry run of A* Search from the start node to the goal node. Show the g(n), h(n) and f(n) values computed at every step, the order in which nodes are expanded, and the final optimal path along with its total cost.

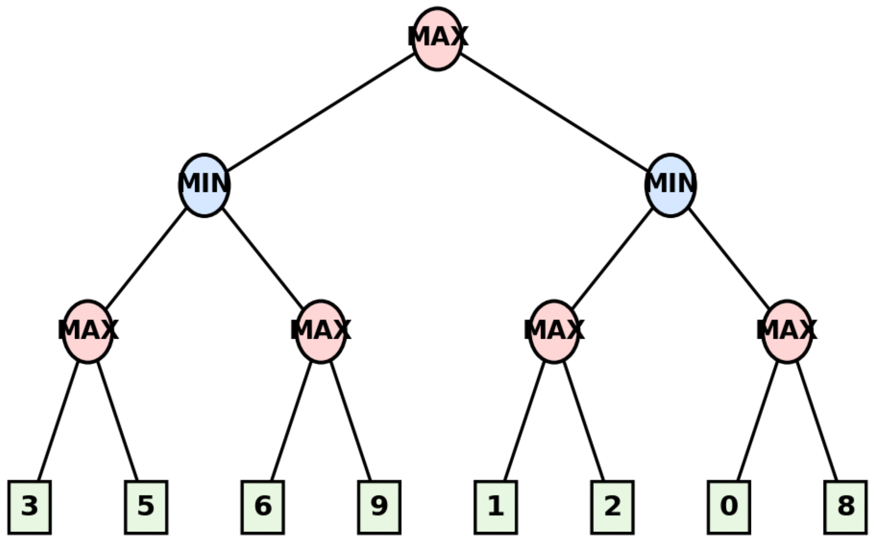


Q3. Formulate a given Constraint Satisfaction Problem (e.g., Map Colouring / N-Queens) by clearly stating the variables, their domains, and the constraints involved. Perform a dry run of the Backtracking Search algorithm, showing each variable assignment, every constraint check, and any backtracking steps, until a complete solution (or failure) is reached.



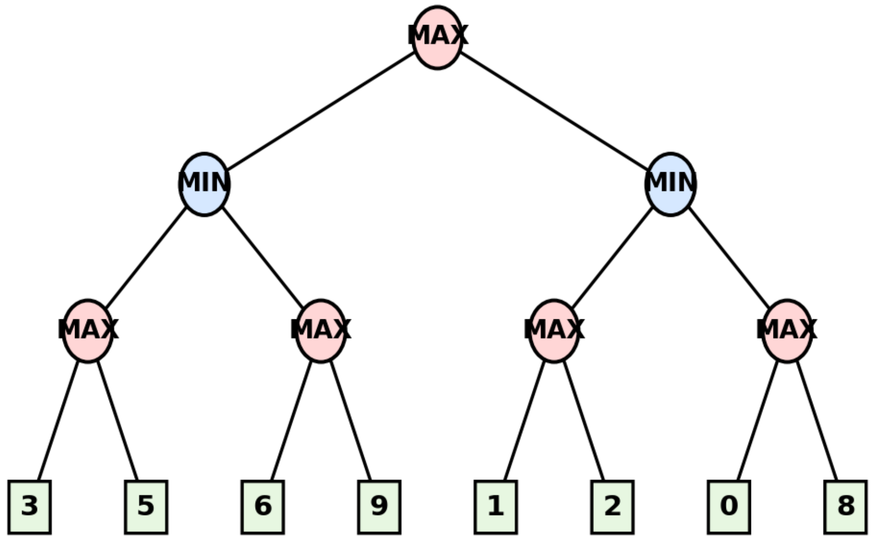
Q4. For a given two-player game tree of your choice (at least 3 levels deep), perform a dry run of the Minimax algorithm. Show the utility values at the leaf/terminal nodes and the backed-up values computed at every internal node, and state the best move for the MAX player with justification.

Root = MAX to move. Squares at the bottom are terminal (leaf) nodes with their utility values.



Q5. Using the same (or a different) game tree from Q4, perform a dry run of the Minimax algorithm with Alpha-Beta Pruning. Show the α and β values maintained at each node during the traversal, clearly indicate which branches get pruned and why, and state the final best move obtained.

Root = MAX to move. Squares at the bottom are terminal (leaf) nodes with their utility values.



Code of Conduct:

I G.Madhan Kumar(192472328)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
G.Madhan Kumar

Dry Run Evaluation

RUBRICS FOR CALCULATION

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Max Marks
Problem Formulation & Setup (states/nodes, heuristic values or CSP variables, domains and constraints correctly identified)	Accurately identifies all required elements (states, heuristics/costs, or variables/domains/constraints) with clear, correct notation.	Identifies most required elements correctly, with only minor omissions or notational errors that do not affect the dry run.	Identifies some required elements; noticeable errors or missing components affect the later steps.	Little or no correct problem formulation; required elements are largely missing or incorrect.	**10**
Correct Application of Algorithm Procedure (expansion order, backtracking, or pruning as per the standard method)	Follows the exact algorithmic procedure at every step with no deviation from the standard method.	Follows the procedure correctly for most steps, with only one or two minor deviations that do not change the outcome.	Several steps deviate from the correct procedure, though the overall approach is recognisable.	Algorithm steps are largely incorrect, or the procedure is not followed.	**10**
Accuracy of Computations (g(n), h(n), f(n) values, minimax backed-up	All computed values are accurate and consistent throughout the dry run.	Minor calculation errors are present but do not significantl	Multiple calculation errors are present that affect the correctness	Calculations are mostly incorrect or missing.	**10**

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Max Marks
values, or α–β updates)		y affect the final outcome.	of intermediate steps.		
Correctness of Final Outcome (final path, CSP solution, or best move obtained)	Final result is completely correct and matches the expected optimal outcome.	Final result is mostly correct, with only a minor deviation from the optimal outcome.	Final result is only partially correct.	Final result is incorrect or not obtained.	**10**
Clarity of Presentation & Justification (trace tables/trees/ diagrams, explanation of each step)	Dry run is presented clearly using well-organised tables/trees/diagrams, with every step explained and justified.	Dry run is reasonably clear and mostly organised, with minor gaps in explanation.	Dry run presentation is unclear or poorly organised, with limited justification.	Dry run is disorganised or illegible, with little to no justification.	**10**
				Total	50

Q1. Dry Run of Greedy Best-First Search
Heuristic Values:

Node	h(n)
S	7
A	6
B	2
C	4
D	3
E	1
G	0

Start Node: S
Goal Node: G
Step 1

- Open List = {S}
- Select S because it is the only node.
- Expand S.
- Add A and B.

Open List = {A(6), B(2)}

Step 2

- Select B because it has the smallest heuristic value (2).
- Expand B.
- Add E.

Open List = {A(6), E(1)}

Step 3

- Select E because h(E)=1.
- Expand E.
- Add G.

Open List = {A(6), G(0)}

Step 4

- Select G because h(G)=0.
- Goal node is reached.

Search stops.

Expansion Order

S → B → E → G

Final Path

S → B → E → G

2.Heuristic Values

Node	h(n)
S	7
A	5
B	2
C	3
D	2
E	1
G	0

Formula:

f(n) = g(n) + h(n)

Step 1

Open List = {S}

Node	g	h	f
S	0	7	7

Expand S

Add A and B.

Step 2

Node	g	h	f
A	2	5	7
B	4	2	6

Select B (lowest f).
Expand B and add E.

Step 3

Node	g	h	f
A	2	5	7
E	6	1	7

Select A.
Expand A and add C and D.

Step 4

Node	g	h	f
C	5	3	8
D	7	2	9
E	6	1	7

Select E.
Expand E and add G.

Step 5

Node	g	h	f
G	7	0	7

Select G.
Goal reached.

Expansion Order
S → B → A → E → G
Final Path
S → B → E → G
Total Cost
4 + 2 + 1 = 7

Q3. Constraint Satisfaction Problem (Map Colouring)
Variables
A, B, C, D
Domain
{Red, Green, Blue}
Backtracking Dry Run
Step 1
Assign A = Red

Step 2
Assign B = Green

Step 3
Try C = Red
Not allowed because A = Red.
Try C = Green
Not allowed because B = Green.
Try C = Blue

Step 4
Assign D = Green
Not allowed because B = Green.
Assign D = Blue
Not allowed because C = Blue.
Assign D = Red

Final Assignment

Variable	Colour
A	Red
B	Green
C	Blue
D	Red

Q4. Dry Run of Minimax Algorithm
Step 1
Leaf values:

- 3
- 5
- 2
- 9

Step 2
Node B is MIN.
Minimum(3,5) = 3
Node C is MIN.
Minimum(2,9) = 2

Step 3
Root A is MAX.
Maximum(3,2) = 3

Best Move
MAX player chooses B.
Q5. Minimax with Alpha-Beta Pruning

Step 1
Start
$\alpha = -\infty$
$\beta = +\infty$

Step 2
Visit node B.
Leaf 3
$\beta = 3$
Leaf 5
Minimum = 3
Return 3.
Root updates
$\alpha = 3$

Step 3
Visit node C.

Current
$\alpha = 3$
$\beta = +\infty$
First leaf = 2
$\beta = 2$
Since $\beta \leq \alpha$ ($2 \leq 3$)
Remaining branch 9 is pruned.